Ex. 2 SQL

unique depart.

Q1 STUDENTS

1. SELECT DISTINCT department
FROM students;

output department
IT
HR
Finance

2. ~~SELECT student_id, name, department~~
~~FROM students~~

average age

2. SELECT department, AVG(age) AS Avg_age
FROM students
GROUP BY department;

output

department	Avg_age
IT	20.5
HR	22
Finance	23

considers all entries in the column.

3. SELECT department, COUNT(*) AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) > 1;

output

department	student_count
IT	2
HR	2

4. SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;

Count will include both 21 & 23

output

student_id	name,	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. SELECT student_id, name, age, department
FROM students
WHERE ~~age~~ department IN ('IT', 'HR')
AND age > 21;

Confines search to these dept

output

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

calculates the total & note that output is grouped by dept.

Q2 COURSES

6. SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;

output

department	total_credits
IT	11

7. SELECT course_id, course_name, department, credits
FROM courses
WHERE credits <> 4;

confines search to the exact #: any course that does not have 4 credits

output

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

output

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

Q3 ENROLLMENT

9. SELECT MAX (grade) AS max-grade,
MIN (grade) AS min-grade,
AVG (grade) AS avg-grade
FROM enrollments;

output

max-grade	min-grade	avg-grade
90	78	84.6

10. SELECT course_id, COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;

output

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

Qt: SALARIES

11. SELECT department,
SUM(salary) AS total_salary,
SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;

output

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12. SELECT department, AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
WHERE AVG(salary) > 55000;

output

department	avg_salary
IT	61000
Finance	70000

13. SELECT employee_id, name, salary, bonus, salary + bonus AS total_compensation
FROM salaries
WHERE salary + bonus > 60000;

output

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Mike	62000	5500	67500

Q5 PROJECTS

14. SELECT department, sum(budget) AS total_budget,
AVG(budget) AS avg-budget
FROM projects
GROUP BY department
HAVING AVG(budget) > 70000;

output

department	total-budget	avg-budget
IT	270000	135000
Finance	80000	80000

15. SELECT department, project_id, project_name, budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000
AND department != 'Marketing';

output

Project_id	Project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	60000
5	HR Portal	HR	50000